



Girraween High School

Year 12 HSC Half Yearly Examination

MATHEMATICS EXTENSION 1

March 2015

Time Allowed: Two hours
(plus 5 minutes reading time)

Instructions To Students

- Attempt all questions.
- All necessary working must be shown for Questions 6 – 10.
- Marks may be deducted for careless or badly arranged work.
- Board approved calculators may be used.
- For Questions 1 -5, write the letter corresponding to the correct answer on your answer sheet.
- For Questions 6 – 10, start each question on a new sheet of paper. Each question should be clearly labelled.

Section I

5 marks

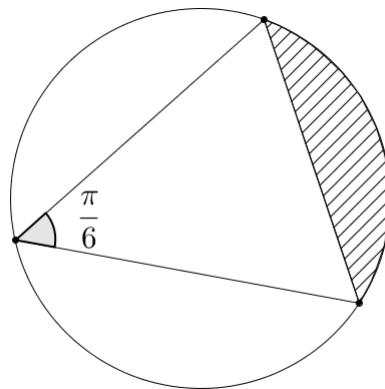
Attempt Questions 1-5

Question 1 (1 mark)

The angle 70° in radians is:

- A. $\frac{7\pi}{18}$
- B. $\frac{18\pi}{7}$
- C. $\frac{7\pi}{36}$
- D. $\frac{36\pi}{7}$

Question 2 (1 mark)



The diagram above is of a unit circle. The shaded area is given by:

- A. $\frac{\pi}{12}$
- B. $\frac{\pi}{6}$
- C. $\frac{\pi - 3}{12}$
- D. $\frac{2\pi - 3\sqrt{3}}{12}$

Question 3 on the next page

Question 3 (1 mark)

A polynomial of degree four is divided by a polynomial of degree two. What is the maximum possible degree of the remainder?

- A. 0
- B. 1
- C. 2
- D. 3

Question 4 (1 mark)

Which is the domain of $f(x) = \ln(x^2 - 1)$?

- A. $x > 0$
- B. $x > 1$
- C. $x < -1$ and $x > 1$
- D. $-1 < x < 1$

Question 5 (1 mark)

The number of different arrangements of the letters of the word REGISTER which begin and end with the letter R is:

- A. $\frac{6!}{(2!)^2}$
- B. $\frac{8!}{2!}$
- C. $\frac{6!}{2!}$
- D. $\frac{8!}{2!2!}$

Question 6 on the next page

Section II

86 marks

Attempt Questions 6-10

Write your answers on the paper provided.

In Questions 6-10, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (23 marks)

- (a) Solve $\frac{x^2 + 5}{x} > 6$. [5]
- (b) Let A be the point $(-2, 7)$ and let B be the point $(1, 5)$. Find the coordinates of the point P which divides the interval AB externally in the ratio $1 : 2$. [3]
- (c) The graphs of the line $x - 2y + 3 = 0$ and the curve $y = x^3 + 1$ intersect at $(1, 2)$. Find the acute angle between the line and the tangent to the curve at the point of intersection. [5]
- (d) The variable point $(3t, 2t^2)$ lies on a parabola. Find the Cartesian equation for this parabola. [2]
- (e) i. Express $\sin x + \sqrt{3} \cos x$ in the form $A \sin(x + \alpha)$. [5]
ii. Hence, or otherwise, solve the equation $\sin x + \sqrt{3} \cos x = \sqrt{3}$ for $0 \leq x \leq 360^\circ$. [3]

The exam continues on the next page

Question 7 (20 marks)

(a) Differentiate:

i. $y = \frac{1}{e^{x^2} + 1}$ [2]

ii. $y = \ln \frac{2}{x}$ [2]

iii. $y = \ln \sqrt{x(1-x)}$ [3]

(b) Consider the function $f(x) = -e^x + 1$.

i. State the domain and range of $f(x)$. [2]

ii. Prove that $f(x)$ is concave down for all x in the domain of $f(x)$, you must show working. [2]

(c) Find:

i. $\int_1^2 \frac{1}{5-2x} dx$ [2]

ii. $\int \frac{4e^{-x}}{1-2e^{-x}} dx$ [2]

(d) i. Solve $\log_2 x = \log_2 \frac{1}{x} + \log_2 (2x-1)$. [3]

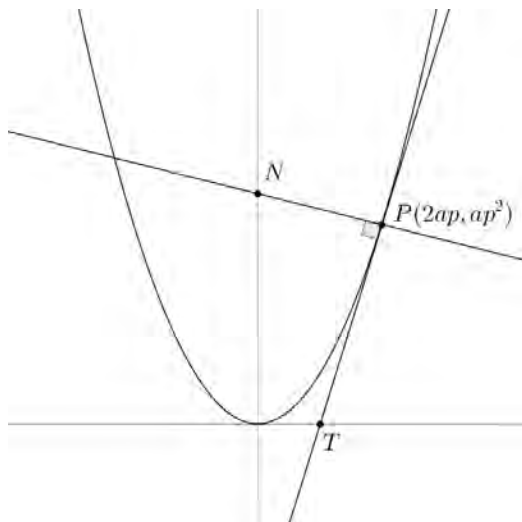
ii. Solve $3^x = 18$. Give your answer to two decimal places. [2]

The exam continues on the next page

Question 8 (14 marks)

- (a) i. Find the remainder obtained by dividing $P(x) = x^3 - bx^2 - bx + 4$ by $Q(x) = x - 2$. [1]
- ii. Hence, or otherwise, find a value of the constant b such that $P(x)$ is divisible by $Q(x)$. [1]
- iii. Find all the roots of $P(x)$ for this value of b . [3]
- (b) Find the constant term in the expansion of $\left(3x^2 + \frac{5}{x^3}\right)^{10}$ [3]
- (c) A particular exam contains 10 multiple choice questions, each with four choices. A student sitting this exam guesses all the answers randomly.
- i. What is the probability that the student scores 50% in this exam? Give your answer to 3 decimal places. [2]
- ii. What is the student's most likely score? [4]

Question 9 (19 marks)

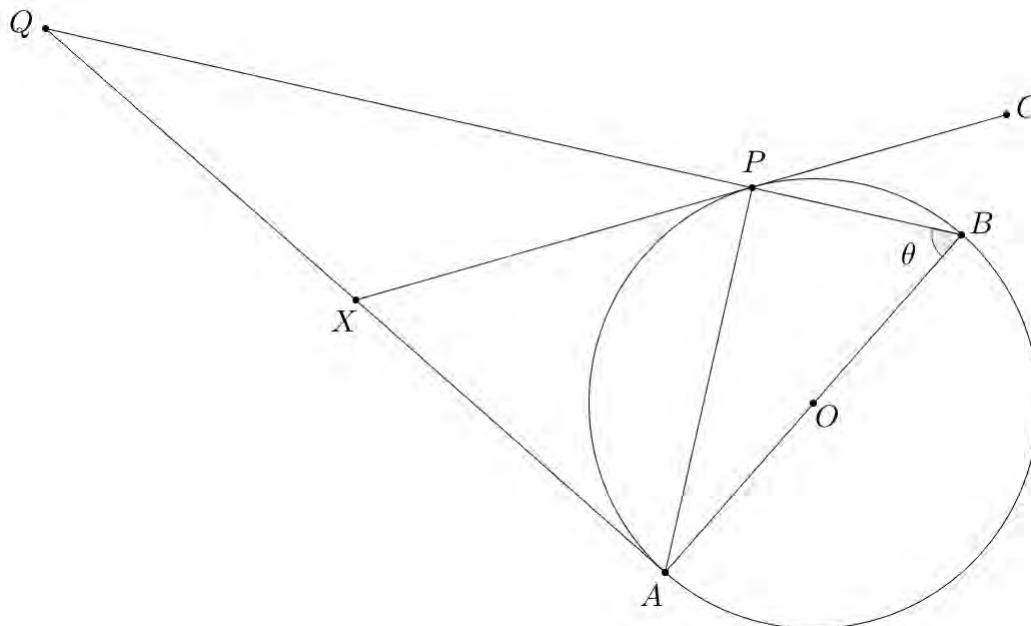


- (a) The diagram shows the graph of the parabola $x^2 = 4ay$. The tangent to the parabola at $P(2ap, ap^2)$ cuts the x -axis at T . The normal to the parabola at P cuts the y -axis at N .
- i. Show that the equation of the tangent at P is $y = px - ap^2$ and find the coordinates of T . [3]
- ii. Show that the coordinates of N are $(0, a(p^2 + 2))$. [2]
- iii. Let M be the midpoint of NT . Show that the locus of M is a parabola and find its focal length. [4]

The exam continues on the next page

Question 9 (continued)

- (b) In the diagram below AB is a diameter of the circle. The tangent AX and chord BP are produced to meet at Q . The tangent CP meets AQ at X .



- i. If $\angle ABP = \theta$, show that $\angle XPQ = 90 - \theta$. [3]
 - ii. Show that X is the midpoint of AQ . [3]
- (c) It can be shown that $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$ for all values of θ . (Do NOT prove this.) [4]

Use this result to solve $\sin 3\theta + \sin 2\theta = \sin \theta$ for $0 \leq \theta \leq 2\pi$.

The exam continues on the next page

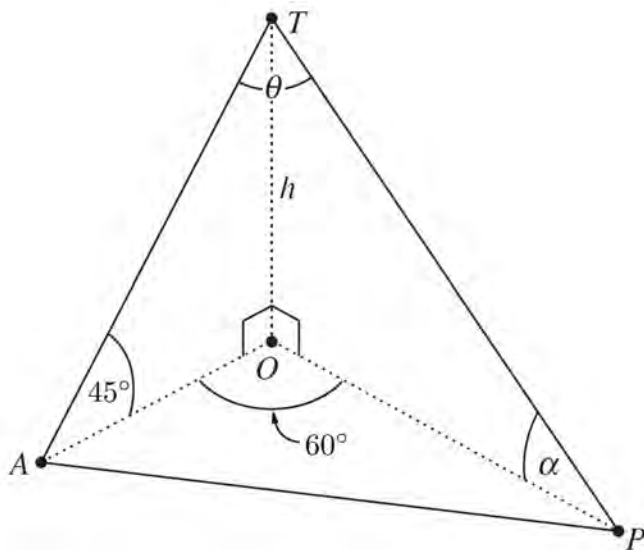
Question 10 (12 marks)

- (a) Use mathematical induction to prove that, for integers $n \geq 1$,

[4]

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \cdots + \frac{n}{(n+1)!} = \frac{(n+1)! - 1}{(n+1)!}$$

- (b) The diagram below shows a vertical tower OT of height h . The angle of elevation of T from A is 45° and $\angle AOP = 60^\circ$. Let the angle of elevation of T from P be α and let $\angle ATP = \theta$.



- i. Show that $OA = h$ and $OP = h \cot \alpha$

[1]

- ii. Show that

[2]

$$AP^2 = h^2 + h^2 \cot^2 \alpha - h^2 \cot \alpha$$

- iii. Using $\triangle ATP$, show that

[2]

$$AP^2 = 3h^2 + h^2 \cot^2 \alpha - 2\sqrt{2}h^2 \operatorname{cosec} \alpha \cos \theta$$

- iv. Show that

[3]

$$\cos \theta = \frac{1}{\sqrt{2}} \sin \alpha + \frac{1}{2\sqrt{2}} \cos \alpha$$

End of exam

Yr12 3U HSC TASK 2 SOLU

MC: A, D, B, C, C

Q1/

$$= 70 \times \frac{\pi}{180} = \frac{7\pi}{18} \therefore \textcircled{A}$$

Q2/

$$A = \frac{1}{2} r^2 (\theta - \sin \theta)$$

but $\theta = \frac{\pi}{3}$ (∠ at centre is double ∠ at circumference).

$$\therefore A = \frac{1}{2} \left(\frac{\pi}{3} - \sin \frac{\pi}{3} \right)$$

$$= \frac{1}{2} \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right)$$

$$= \frac{1}{2} \left(\frac{2\pi - 3\sqrt{3}}{6} \right)$$

$$= \frac{2\pi - 3\sqrt{3}}{12} \therefore \textcircled{D}$$

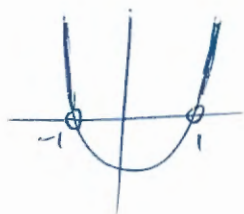
Q3/

ⓑ

Q4/

$$x^2 - 1 > 0$$

$$(x-1)(x+1) > 0$$



$\therefore D: x < -1 \text{ \& } x > 1 \therefore \textcircled{C}$

Q5/

REGISTER.

$$= \frac{6!}{2!} \therefore \textcircled{C}$$

Q6/

$$(1) \frac{n^2 + 5}{n} > 6$$

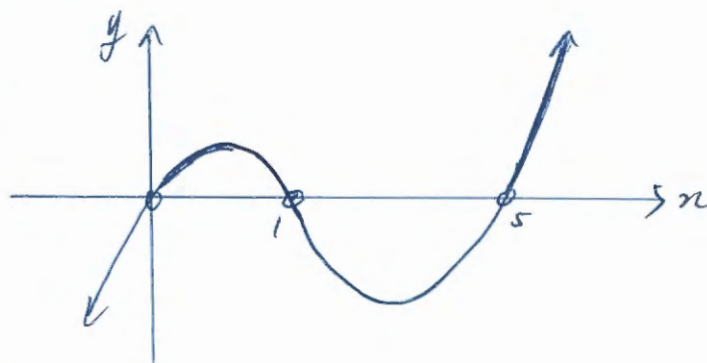
$$n(n^2 + 5) > 6n^2$$

$$n(n^2 + 5) - 6n^2 > 0$$

$$n[(n^2 + 5) - 6n] > 0$$

$$n(n^2 - 6n + 5) > 0$$

$$n(n-5)(n-1) > 0$$



$$\therefore 0 < n < 1 \text{ \& } n > 5$$

(b)

$$(-2, 7) \quad (1, 5)$$

$$\begin{array}{c} \times \\ -1 : 2 \end{array}$$

$$x = \frac{-1-4}{1} \quad y = \frac{-5+14}{1}$$

$$= -5 \quad = 9$$

$$\therefore P = (-5, 9)$$

Q6/

(c) $y = x^3 + 1$

$$y' = 3x^2$$

$$y'(1) = 3$$

$$2y = x + 3$$

$$y = \frac{1}{2}x + \frac{3}{2}$$

$$m_1 = 3 \quad m_2 = \frac{1}{2}$$

$$\tan \theta = \left| \frac{3 - \frac{1}{2}}{1 + \frac{3}{2}} \right|$$

$$= \left| \frac{6 - 1}{2 + 3} \right| = \left| \frac{5}{5} \right| = 1$$

$$\therefore \theta = 45^\circ$$

(d) $x = 3t$ & $y = 2t^2$

$$t = \frac{x}{3}$$

$$\therefore y = 2\left(\frac{x}{3}\right)^2 = \frac{2}{9}x^2$$

$$\therefore y = \frac{2}{9}x^2$$

(e)

(i)

$$\sin x + \sqrt{3} \cos x = A \sin(x + \alpha)$$

$$= A [\sin x \cos \alpha + \cos x \sin \alpha]$$

$$= A \cos \alpha \sin x + A \sin \alpha \cos x$$

$$\therefore A \cos \alpha = 1 \quad \& \quad A \sin \alpha = \sqrt{3}$$

$$A^2 \cos^2 \alpha + A^2 \sin^2 \alpha = 1 + 3 = 4$$

$$\therefore A^2 = 4 \quad \therefore A = 2$$

$$\therefore \cos \alpha = \frac{1}{2} \quad \therefore \alpha = 60^\circ$$

$$\therefore \sin x + \sqrt{3} \cos x = 2 \sin(x + 60^\circ)$$

(ii)

$$2 \sin(x + 60^\circ) = \sqrt{3}$$

$$\sin(x + 60^\circ) = \frac{\sqrt{3}}{2}$$

$$\therefore x + 60^\circ = 60^\circ \text{ or } 120^\circ$$

$$\text{But } 0 \leq x \leq 360$$

$$\therefore 60 \leq x + 60 \leq 420$$

$$\therefore x + 60^\circ = 60^\circ, 120^\circ, 420^\circ$$

$$\therefore x = 0^\circ, 60^\circ, 360^\circ$$

Q7

(a)

$$(i) f = \frac{1}{e^{x^2} + 1}$$

$$f = (e^{x^2} + 1)^{-1}$$

$$f' = -(e^{x^2} + 1)^{-2} \times (2xe^{x^2})$$

$$f' = \frac{-2xe^{x^2}}{(e^{x^2} + 1)^2}$$

$$(ii) f = \ln\left(\frac{2}{x}\right)$$

$$f = \ln 2 - \ln x$$

$$f' = -\frac{1}{x}$$

$$(iii) f = \ln(x(1-x))^{\frac{1}{2}}$$

$$f = \frac{1}{2} \ln x(1-x)$$

$$f = \frac{1}{2} \left[\ln x + \ln(1-x) \right]$$

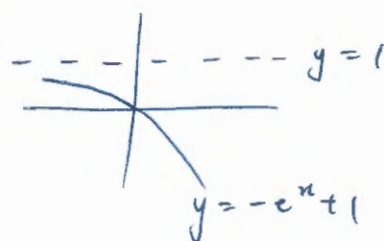
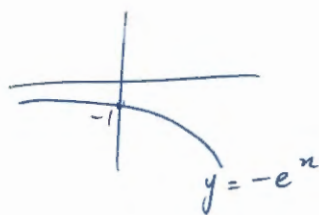
$$f' = \frac{1}{2} \left[\frac{1}{x} - \frac{1}{1-x} \right]$$

$$f' = \frac{1}{2} \left[\frac{1-x-x}{x(1-x)} \right]$$

$$f' = \frac{1-2x}{2x(1-x)}$$

(b)

(i) D: all real x



$$\therefore R: y < 1$$

$$(ii) y = -e^x + 1$$

$$y' = -e^x$$

$$y'' = -e^x$$

since $e^x > 0$ for all x

$$\therefore -e^x < 0 \text{ for all } x$$

$\therefore f(x)$ is concave down for all x .

Q7

$$(c) \quad (i) \int_1^2 \frac{1}{5-2u} du$$

$$= -\frac{1}{2} \int_1^2 \frac{-2}{5-2u} du$$

$$= -\frac{1}{2} [\ln(5-2u)]_1^2$$

$$= -\frac{1}{2} [\ln 1 - \ln 3]$$

$$= \frac{1}{2} \ln 3$$

$$(ii) \int \frac{4e^{-u}}{1-2e^{-u}} du$$

$$= 2 \int \frac{2e^{-u}}{1-2e^{-u}} du$$

$$= 2 \ln(1-2e^{-u}) + C$$

(d)

$$(i) \log_2 u = \log_2 \frac{1}{u} + \log_2 (2u-1)$$

$$\log_2 u = -\log_2 u + \log_2 (2u-1)$$

$$2\log_2 u = \log_2 (2u-1)$$

$$\log_2 u^2 = \log_2 (2u-1)$$

$$\therefore u^2 = 2u-1$$

$$u^2 - 2u + 1 = 0$$

$$\therefore (u-1)^2 = 0 \quad \therefore u = 1$$

$$(ii) 3^x = 18$$

$$x = \log_3 18$$

$$x = \frac{\ln 18}{\ln 3} = 2.63 \text{ (2dp)}$$

Q8

(a)

$$(i) r = p(2)$$

$$r = 8 - 4b - 2b + 4$$

$$r = 12 - 6b$$

$$(ii) b = 2$$

(iii)

$$p(u) = u^3 - 2u^2 - 2u + 4$$

$$\begin{array}{r} u^2 - 2 \\ u - 2 \overline{) u^3 - 2u^2 - 2u + 4} \\ \underline{-(u^3 - 2u^2)} \\ -2u + 4 \\ \underline{-(-2u + 4)} \\ 0 \end{array}$$

$$\therefore p(u) = (u-2)(u^2-2)$$

$$= (u-2)(u-\sqrt{2})(u+\sqrt{2})$$

$$\therefore \text{roots are: } 2, \pm\sqrt{2}$$

Q8

$$(b) \left(3n^2 + \frac{5}{n^3}\right)^{10}$$

$$\begin{aligned} T_{k+1} &= \binom{10}{k} (3n^2)^{10-k} \left(\frac{5}{n^3}\right)^k \\ &= \binom{10}{k} 3^{10-k} n^{20-2k} 5^k n^{-3k} \\ &= \binom{10}{k} 3^{10-k} 5^k n^{20-5k} \end{aligned}$$

For constant term: $20 - 5k = 0$
 $\therefore k = 4$

$\therefore$ Constant term is:

$$T_5 = \binom{10}{4} 3^6 5^4$$

(c)

(i) Let X denote number of correct answers.

$$\begin{aligned} P(X=5) &= \binom{10}{5} \left(\frac{3}{4}\right)^5 \left(\frac{1}{4}\right)^5 \\ &= 0.058 \text{ (3dp)} \end{aligned}$$

(ii)

$$\frac{P_{k+1}}{P_k} = \frac{\binom{10}{k} \left(\frac{3}{4}\right)^{10-k} \left(\frac{1}{4}\right)^k}{\binom{10}{k-1} \left(\frac{3}{4}\right)^{11-k} \left(\frac{1}{4}\right)^{k-1}}$$

$$= \frac{10!}{(10-k)! k!} \times \frac{(11-k)! (k-1)!}{10!} \times \frac{4}{3} \times \frac{1}{4}$$

$$= \frac{11-k}{k} \times \frac{1}{3} = \frac{11-k}{3k}$$

$$\frac{11-k}{3k} \geq 1 \text{ for greatest probability}$$

$$11-k \geq 3k$$

$$4k \leq 11$$

$$k \leq \frac{11}{4} = 2.75$$

$$\therefore k = 2$$

$$\therefore \text{Most likely score is } \frac{2}{10} = 20\%$$

Q9

(a)

$$(i) x^2 = 4ay$$

$$\text{So } y = \frac{1}{4a} x^2$$

$$y' = \frac{1}{2a} x$$

$$y'(2ap) = p$$

$$y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = pn - 2ap^2$$

$$y = pn - ap^2$$

$$y = 0 \text{ when } pn - ap^2 = 0$$

$$p(n - ap) = 0$$

$$\therefore n = ap$$

$$\therefore T = (ap, 0)$$

(a)

$$y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$x=0$ when

$$y - ap^2 = -\frac{1}{p} (-2ap)$$

$$g - ap^2 = 2a$$

$$g = 2a + ap^2 = a(2 + p^2)$$

$$\therefore N = (0, a(p^2 + 2))$$

(iii)

$$M = \left(\frac{ap+0}{2}, \frac{a(p^2+2)+0}{2} \right)$$

$$m = \left(\frac{1}{2}ap, \frac{1}{2}a(p^2+2) \right)$$

$$\therefore x = \frac{1}{2} a y \quad y = \frac{1}{2} a (p^2 + 2)$$

$$\text{So } p = \frac{2\pi}{9}$$

$$\text{So } y = \frac{1}{2}a \left(\frac{4x^2}{a^2} + 2 \right)$$

$$y = \frac{2x^2}{a} + a$$

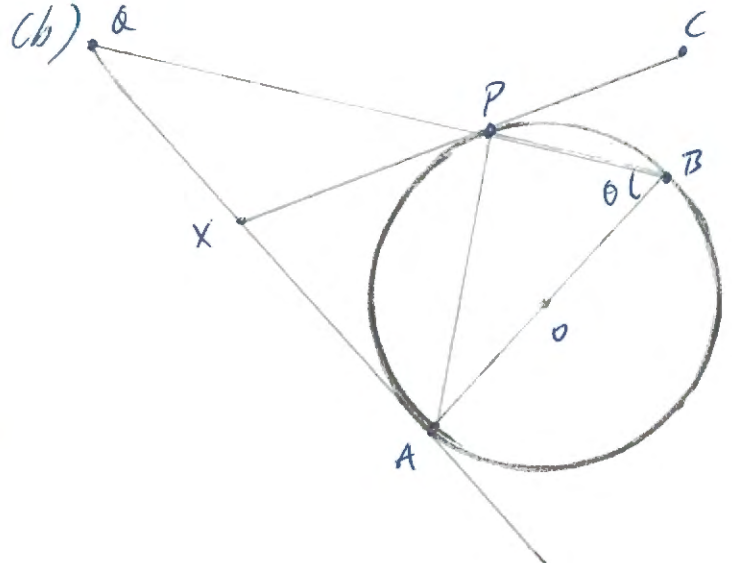
$$y - a = \frac{2n^2}{a}$$

$$\frac{a}{2}(y-a) = x^2$$

$$4 \times (\text{focal length}) = \frac{9}{2}$$

focal length
is $\frac{u}{8}$

(b) $\mathbb{Q}$



(i) $\angle APB = 90^\circ$ ($\angle$ in semi-circle with diameter AB)

$\angle XPA = \theta$ ($\angle$ between tangent XP and chord AP equals $\angle$ in the alternate segment).

$$\angle X P Q = 90^\circ - \theta \quad (\angle \text{sum of straight line } Q P B)$$

(iii)

($\therefore$) $\angle QAB = 90^\circ$ (tangent $\perp$ to radius at point of contact)

$\therefore \angle AQP = 90^\circ - \theta$ ($\angle$ sum of $\triangle ABQ$)

$\therefore QX = XP$ (equal sides
opposite equal $\angle$'s
of $\triangle XQP$)

Q9

(b)

(ii) $XP = XA$ (tangents from an external point)

$\therefore AX = XA \therefore X$ is midpoint of QA .

(c)

$$\sin 3\theta + \sin 2\theta = \sin \theta$$

$$3\sin \theta - 4\sin^3 \theta + 2\sin \theta \cos \theta = \sin \theta$$

$$4\sin^3 \theta - 2\sin \theta - 2\sin \theta \cos \theta = 0$$

$$\sin \theta [4\sin^2 \theta - 2\cos \theta - 2] = 0$$

$$\sin \theta [4(1 - \cos^2 \theta) - 2\cos \theta - 2] = 0$$

$$\sin \theta [4 - 4\cos^2 \theta - 2\cos \theta - 2] = 0$$

$$\sin \theta [-4\cos^2 \theta - 2\cos \theta + 2] = 0$$

$$-\sin \theta [4\cos^2 \theta + 2\cos \theta - 2] = 0$$

$$-\sin \theta (2\cos \theta - 1)(2\cos \theta + 2) = 0$$

$$\therefore \sin \theta = 0, \cos \theta = \frac{1}{2}, \cos \theta = -1$$

$$\therefore \theta = 0, \pi, 2\pi, \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\theta = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$$

Q10

(a)

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = \frac{(n+1)! - 1}{(n+1)!}$$

Prove true for $n=1$:

$$LHS = \frac{1}{2!} = \frac{1}{2}$$

$$RHS = \frac{2! - 1}{2!} = \frac{1}{2!} = \frac{1}{2}$$

$\therefore LHS = RHS \therefore$ true for $n=1$.

Assume true for $n=k$, i.e.

$$\frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k}{(k+1)!} = \frac{(k+1)! - 1}{(k+1)!}$$

Prove true for $n=k+1$:

$$\frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!}$$

$$= \frac{(k+2)! - 1}{(k+2)!}$$

$$LHS = \frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!}$$

$$= \frac{(k+1)! - 1}{(k+1)!} + \frac{k+1}{(k+2)!}$$

$$= \frac{(k+2)! - (k+2) + k+1}{(k+2)!}$$

$$= \frac{(k+2)! - k - 2 + k + 1}{(k+2)!}$$

$$= \frac{(k+2)! - 1}{(k+2)!} = RHS.$$

$\therefore$ true for $n=k+1$

$\therefore$ By induction, it is true for all $n \geq 1$.

Q10

(b)

(i) In ΔATO :

$$\tan 45 = \frac{h}{OA}$$

$$\therefore OA = h$$

In ΔTOP :

$$\tan \alpha = \frac{h}{OP}$$

$$\therefore OP = h \cot \alpha$$

(ii) Using cosine rule on ΔAOP :

$$AP^2 = OA^2 + OP^2 - 2 \times OA \times OP \times \cos 60$$

$$AP^2 = h^2 + h^2 \cot^2 \alpha - 2h^2 \cot \alpha \times \frac{1}{2}$$

$$AP^2 = h^2 + h^2 \cot^2 \alpha - h^2 \cot \alpha$$

(iii)

$$AT^2 = h^2 + h^2 = 2h^2$$

$$\therefore AT = \sqrt{2}h$$

$$\begin{aligned} TP^2 &= h^2 + h^2 \cot^2 \alpha \\ &= h^2 (1 + \cot^2 \alpha) \end{aligned}$$

$$\begin{aligned} \therefore TP &= h \sqrt{1 + \cot^2 \alpha} \\ &= h \operatorname{cosec} \alpha \end{aligned}$$

(iii)

So:

$$\begin{aligned} AP^2 &= AT^2 + TP^2 - 2 \times AT \times TP \times \cos \theta \\ &= 2h^2 + h^2 + h^2 \cot^2 \alpha - 2\sqrt{2}h^2 \operatorname{cosec} \alpha \cos \theta \\ &= 3h^2 + h^2 \cot^2 \alpha - 2\sqrt{2}h^2 \operatorname{cosec} \alpha \cos \theta \end{aligned}$$

(iv)

Equating the expressions for AP^2 from part (ii) & (iii) we get:

$$\begin{aligned} h^2 + h^2 \cot^2 \alpha - h^2 \cot \alpha &= \\ 3h^2 + h^2 \cot^2 \alpha - 2\sqrt{2}h^2 \operatorname{cosec} \alpha \cos \theta \end{aligned}$$

$$2h^2 + h^2 \cot^2 \alpha - 2\sqrt{2}h^2 \operatorname{cosec} \alpha \cos \theta = 0$$

$$2 + \cot^2 \alpha - 2\sqrt{2} \operatorname{cosec} \alpha \cos \theta = 0$$

$$2 + \frac{\cos^2 \alpha}{\sin^2 \alpha} - 2\sqrt{2} \frac{\cos \theta}{\sin \alpha} = 0$$

$$2 \sin^2 \alpha + \cos^2 \alpha - 2\sqrt{2} \cos \theta = 0$$

$$2\sqrt{2} \cos \theta = 2 \sin^2 \alpha + \cos^2 \alpha$$

$$\cos \theta = \frac{1}{\sqrt{2}} \sin^2 \alpha + \frac{1}{2\sqrt{2}} \cos^2 \alpha$$